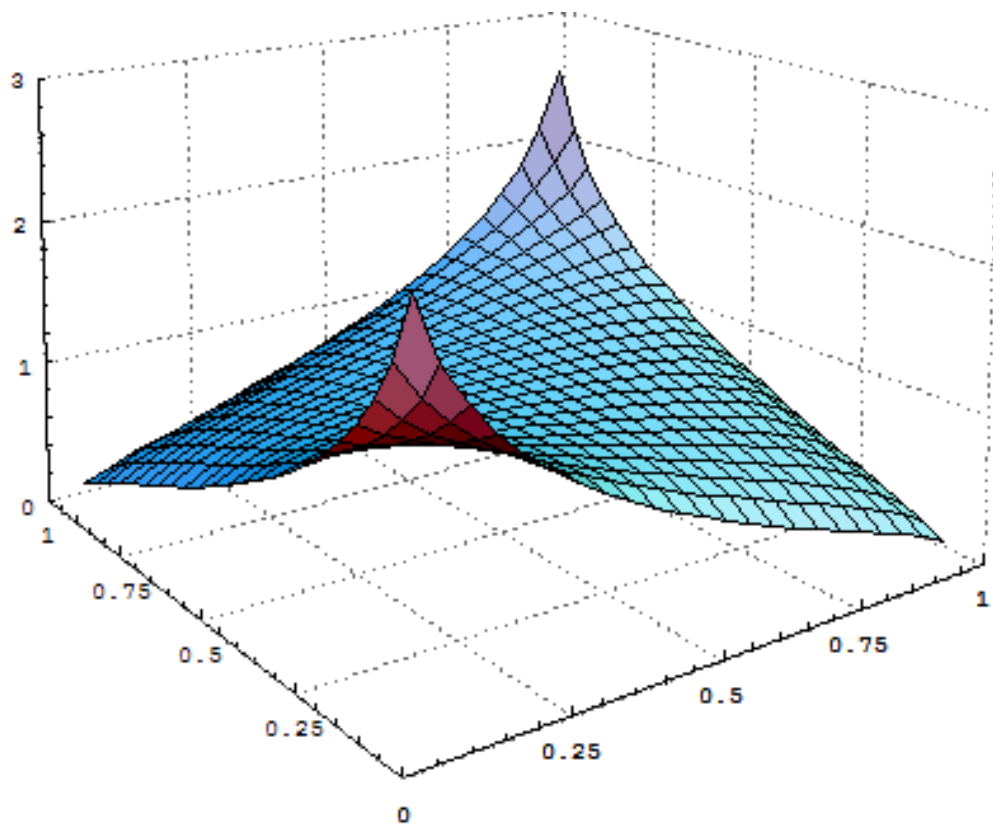




A Theoretical Dive into Copulas

STAT 590 Final Project

Submitted to

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I. Introduction

A copula (from the Latin *copulare*, meaning “to bind”, “unite”, or “join”) is a multivariate cumulative distribution function that helps us fully explain the dependence structure between two random variables. An important aspect of the copula is that the marginal probability distribution for each of its variables follows a uniform distribution on the interval $[0,1]$. Therefore, instead of modeling raw values of data, the copula models how percentiles move together – thus creating a universal scale completely independent of how the original variables are distributed. Copulas are groundbreaking because they provide a succinct way to fully explain a joint distribution by splitting it into its marginals and its dependence structure (see Sklar’s Theorem in Background).

The copula also allows for applications of correlation between two variables, mainly the idea of concordance. Concordance is a metric that describes how variables are ordered amongst themselves. For example, two independent observations of two variables are considered concordant if they are both ordered in the same way (see Concordance Index in Background). The idea of concordance can be applied to 2 other measures: Kendall’s tau and the Concordance index (or C-index). Both measures are tightly coupled, and they provide a fantastic way of measuring whether pairs of observations are ordered in the same way.

From copulas, Kendall’s tau and the C-index can be derived. It is important to know the underlying theory within these measures and how to derive them from various copulas. Therefore, in the proceeding sections, we will explore these different copulas and their corresponding concordance metrics.

II. Background

Sklar's Theorem

Sklar was a mathematician of the mid 20th century who introduced the notion of copulas into probability theory. Sklar, a second-generation Ukrainian immigrant, was born in Chicago in 1925. At the young age of 16, he attended the University of Chicago, and he earned his PhD at the California Institute of Technology in 1956. Three years later, in 1959, he birthed copulas into the world and proved his famous theorem: Sklar's Theorem. Sklar’s Theorem states the following:

For any multivariate CDF $H(x_1, x_2)$ with continuous marginals $F_1(x_1)$ and $F_2(x_2)$ there exists a unique copula C such that

$$H(x_1, x_2) = C(F_1(x_1), F_2(x_2)) \quad (1.1a)$$

Define $u = F_X(x)$ and $v = F_Y(y)$. The copula equivalently can be written:

$$C(u, v) = P(U \leq u, V \leq v) \quad (1.1b)$$

Where $U = F_X(X)$ and $V = F_Y(Y)$, the probability integral transform.

Sklar's theorem is the backbone of copula theory. From it, a plethora of copula can be derived. This theorem is extremely powerful not only for its ability to derive copulas, but also because it connects a joint distribution perfectly to its marginals while retaining all information.

Kendall's Tau

Kendall was a prominent British Statistician born in Kettering, Northamptonshire. He suffered from cerebral meningitis as a child yet went on to survive and study Mathematics at Cambridge. He worked on his famous tau in the mid 20th century, leading to a monograph on Rank Correlation in 1948. Kendall's tau can be derived as follows:

Let (X_1, Y_1) and (X_2, Y_2) be two independent sets of $(X, Y) \sim^{iid} F_{x,y}(x, y), F_x(x), F_y(y)$. If $(X_1 - Y_1)(X_2 - Y_2) > 0$, then these sets are **concordant**. Similarly, If $(X_1 - Y_1)(X_2 - Y_2) < 0$, then these sets are **discordant**. Kendall's tau is thus the probability of concordance minus the probability of discordance:

$$\tau = P(\text{concordance}) - P(\text{discordance}) \quad -1 \leq \tau \leq 1 \quad (1.2)$$

We can define this τ more mathematically as such:

Define the random event $A = \{X_1 < X_2, Y_1 < Y_2\}$, $B = \{X_1 > X_2, Y_1 > Y_2\}$, $C = \{X_1 < X_2, Y_1 > Y_2\}$, and $D = \{X_1 > X_2, Y_1 < Y_2\}$. It can be shown that

$$P(\text{concordance}) = P(X_1 - X_2 < 0, Y_1 - Y_2 < 0) + P(X_1 - X_2 > 0, Y_1 - Y_2 > 0) = P(A) + P(B)$$

$$P(\text{discordance}) = P(X_1 - X_2 < 0, Y_1 - Y_2 > 0) + P(X_1 - X_2 > 0, Y_1 - Y_2 < 0) = P(C) + P(D)$$

Since (X, Y) is iid:

$$P(A) = P(B)$$

and

$$P(C) = P(D)$$

Therefore, rewriting Kendall's tau:

$$\tau = 2P(A) - 2P(C)$$

Finally, (X, Y) is continuous, and we are not accounting for ties and thus:

$$P(A) + P(C) = \frac{1}{2}$$

and further,

$$P(C) = \frac{1}{2} - P(A)$$

so

$$\tau = 4P(A) - 1$$

Converting back to the random event A:

$$\tau = 4P(X_1 < X_2, Y_1 < Y_2) - 1$$

Finally, consider:

$$P(A|X_2 = x, Y_2 = y)$$

on support $[0,1]$. Then, this probability can be converted to:

$$P(X_1 < x, Y_2 < y) = C(x, y)$$

Where C is a copula. This follows from the definition of a multivariate cumulative distribution function, which is how a copula was defined (see Equation 1.1). The probability of A can be defined as the expectation of the indicator function of A, and the resulting conclusion can be made:

$$\tau = 4 \int_0^1 \int_0^1 C(u, v) dC(u, v) - 1 \quad (1.3)$$

This important derivation allows Kendall's tau to be derived from the copula. Using this definition, we can now find something very useful called the *concordance index*.

Concordance Index

The concordance index is a measure that describes the concordance of a dataset or distribution. There are many different definitions for the concordance index, but for the sake of this report, I will define the concordance index as:

$$\kappa = P(\text{Concordance}) \quad (1.4)$$

Therefore, the concordance index simply is this: if I were to choose a random independent pair (X_i, Y_i) and compare it to another random independent pair (X_j, Y_j) the probability that these pairs are concordant (i.e. have the same ordering) is equal to κ . Conveniently, κ can be found in terms of τ and therefore, each unique copula will have not only a unique τ but also a unique κ . The derivation is as follows:

Let $\kappa = P(\text{Concordance})$ and $\delta = P(\text{Discordance})$. Then, Kendall's tau can be defined as:

$$\tau = \kappa - \delta$$

Or equivalently,

$$\delta = \kappa - \tau. \quad (1.5)$$

Kendall's tau can also be defined as:

$$\tau = \frac{\kappa - \delta}{\kappa + \delta}$$

Simplifying this expression in terms of κ yields:

$$\kappa = \frac{-\delta(1 + \tau)}{\tau - 1}$$

Then, substituting in Equation 1.5:

$$\kappa = \frac{-(\tau - \kappa)(1 + \tau)}{\tau - 1}$$

Solving for κ yields the final form:

$$\kappa = \frac{\tau+1}{2} \quad 0 \leq \kappa \leq 1 \quad (1.6)$$

Or, in terms of just the Copula, substituting Equation (1.3) for τ :

$$\kappa = 2 \int_0^1 \int_0^1 C(u, v) dC(u, v) \quad (1.7)$$

Now, the three concepts have been connected: the copula, Kendall's tau, and the concordance index. The copula is a function that describes the dependence structure between two random variables X and Y , separate from their marginal distributions. Kendall's tau summarizes this dependence through the ordering of independent pairs, measuring the difference between the probabilities of concordance and discordance. The concordance index expresses this same rank-ordering relationship as a probability because for continuous variables each copula's Kendall's tau can be directly converted into a concordance index. Therefore, deriving copulas and computing their Kendall's taus and concordance indices provides a useful way to compare the strength and direction of rank dependence implied by different copula families. However, these summaries do not capture every feature of a copula, so additional properties of a copula such as tail dependence or asymmetry may also be needed when deciding which copula is most useful for modeling.

III. Derivations

This section will contain derivations for 6 different copulas. The introduced copulas will be derived as well as their specific Kendall's tau and Concordance index. In some cases, particularly in the later cases, I will also explain the dependence parameter of the copula and derive the MLE of the dependence parameter, $\widehat{\theta}_{MLE}$.

Independence Copula

Derivation

The independence copula is one of the simplest and most intuitive copulas. It is as the name suggests: the independence copula models the dependence structure of two random variables that are completely independent from one another. The independence copula is also the first copula in this report considered what is called an *Archimedean copula*. The Archimedean copulas are a popular class of copula for one reason: most Archimedean copulas can admit an explicit formula in terms of a parameter θ , called the *dependence parameter*. Therefore, this class of copula allows for dependence modeling in high dimensional data – meaning the Archimedean class can be applied very easily to real world data. A copula is considered Archimedean if it follows this definition:

Let $\phi: [0,1] \times \Theta \rightarrow [0,\infty)$ be a continuous, strictly decreasing convex function called the generator function such that $\phi(1;\theta) = 0$, $\theta \in \Theta$ and its pseudo-inverse defined by:

$$\phi^{-1}(t;\theta) = \{\phi^{-1}(t;\theta) \text{ if } 0 \leq t \leq \phi(0;\theta), 0 \text{ else}\}$$

Then, we can say that a d dimensional copula C is Archimedean if it can be defined as:

$$C(u_1, \dots, u_d; \theta) = \phi^{-1}(\phi(u_1; \theta) + \dots + \phi(u_d; \theta); \theta) \quad (2.1)$$

If and only if ϕ^{-1} is d -monotone on $[0, \infty)$ and thus differentiable $d - 2$ times where:

$$(-1)^k \phi^{-1, (k)}(t; \theta) \geq 0$$

For all $t \geq 0$ where $k = 0, 1, \dots, d - 2$ and $(-1)^{d-2} \phi^{-1, (d-2)}(t; \theta)$ is strictly nonincreasing and convex.

While this definition is nice, it is hard to define exactly what a generator function will be and how to even go about finding one. Thankfully, most generators can be found very easily by taking the Laplace transform of a non-negative random variable. Doing this will give the pseudo-inverse generator, and thus finding the generator is as simple as taking the inverse of the pseudo-inverse. The Laplace transform can be defined as:

$$\mathcal{L} = E[e^{-tS}] = \phi^{-1}(t; \theta) \quad (2.2)$$

where S is a non-negative random variable with support $S \geq 0$. This is identical to the definition of the moment generating function, but with a negative sign in the exponent.

All the pieces are in place to derive the independence copula and the rest of the Archimedean copulas going forward. The derivation is as follows:

Let S be a degenerate distribution where $S \equiv 1$ and let $\phi_I^{-1}(t) = E[e^{-tS}]$. Since S is degenerate and equal to 1: $\phi_I^{-1}(t) = E[e^{-t \cdot 1}] = e^{-t}$

Taking the inverse of this function yields the generator function:

$$\phi_I(t) = -\ln(t)$$

For simplicity, suppose independent random variables $U \sim Unif[0,1]$ and $V \sim Unif[0,1]$ where $U = F_X(X)$ and $V = F_Y(Y)$ are the probability integral transforms of two continuous random variables X and Y . By equation (2.1), the independence copula can be defined as follows:

$$C(u, v) = \phi_I^{-1}(\phi_I(u) + \phi_I(v)) = \exp\{-(-\ln(u) - \ln(v))\}$$

$$C(u, v) = uv \quad (2.3a)$$

From definition (1.1b), this copula models independence.

$$C(u, v) = P(U \leq u, V \leq v) = P(U \leq u) \cdot P(V \leq v) = uv$$

$$\text{And } P(U \leq u) = u \text{ and } P(V \leq v) = v$$

This is the definition of independence in probability theory.

Kendall's Tau

From the Copula definition and (1.3), Kendall's tau can be derived:

$$\tau = 4 \int_0^1 \int_0^1 C(u, v) dC(u, v) - 1$$

$$\tau = 4 \int_0^1 \int_0^1 uv du dv - 1$$

$$\tau = 4 \int_0^1 \frac{v}{2} dv - 1$$

$$\tau = 4 \left(\frac{1}{4} \right) - 1$$

$$\tau_I = 0 \quad (2.3b)$$

Kendall's tau for the independence copula is 0, meaning the probability of concordance and discordance are both equal. This makes sense – if the variables are truly independent, their ranks will have no effect on one another.

Concordance Index

From the Copula definition and (1.7), the concordance index can be derived:

$$\kappa = 2 \int_0^1 \int_0^1 C(u, v) dC(u, v)$$

$$\kappa = 2 \int_0^1 \int_0^1 uv du dv$$

$$\kappa = 2 \int_0^1 \frac{v}{2} dv$$

$$\kappa = 2 \cdot \left(\frac{1}{4} \right)$$

$$\kappa_I = \frac{1}{2} \quad (2.3c)$$

The concordance index for the independence copula is a constant, $\frac{1}{2}$. This means that the probability of a random pair being concordant is simply 0.5. The independence copula is sometimes considered the *baseline copula*. This is because it serves as a great starting point to start comparing with other copulas, some of which will be derived later.

Fréchet-Hoeffding Bounds

Derivation

The Fréchet-Hoeffding bounds are two distinct copulas that define the bounds that any copula must be in-between. As the name implies, there is a Lower Fréchet-Hoeffding bound and an Upper Fréchet-Hoeffding bound. Since a copula explains how two variables move together in rank, the Lower Fréchet-Hoeffding models the situation in which the relationship between two continuous random variables is perfectly negative, meaning they have opposite rank order. In contrast, the Upper Fréchet-Hoeffding bound describes the situation in which the relationship between two continuous random variables is perfectly positive, meaning they have identical rank order. These copulas are not Archimedean in nature, so they must be found purely with probabilities:

Start with definition (1.1b):

$$C(u, v) = P(U \leq u, V \leq v)$$

Where U and V are the same random variables defined when deriving the independence copula. Consider the upper bound. This is the situation in which the relationship between the two random variables is perfectly positive. This means that the rank order is identical. Therefore, because U and V are independent, we can claim:

$$P(U \leq u, V \leq v) = P(U \leq u) \cdot P(V \leq v) \leq P(U \leq u)$$

And

$$P(U \leq u, V \leq v) = P(U \leq u) \cdot P(V \leq v) \leq P(V \leq v)$$

Since rank order is identical, that means that $U = V$. Thus:

$$P(U \leq u, V \leq v) = P(U \leq u, U \leq v) = \min(u, v)$$

Therefore, the Upper Fréchet-Hoeffding Bound is:

$$C(u, v) = \min(u, v) \quad (2.4a)$$

The Lower Fréchet-Hoeffding Bound can be found similarly by starting with definition (1.1b) and the inclusion-exclusion probability identity:

$$C(u, v) = P(U \leq u, V \leq v)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Now, let $A = U \leq u$ and $B = V \leq v$. Thus, $P(A) = u$ and $P(B) = v$. Change the inclusion-exclusion probability identity to an equivalent inequality:

$$P(A \cup B) \leq 1$$

Thus

$$u + v - P(A \cap B) \leq 1$$

And since $P(A \cap B)$ is just $P(U \leq u, V \leq v)$ because U and V are independent, then

$$C(u, v) = P(A \cap B) = u + v - 1$$

And finally, accounting for the case that $u + v - 1 < 0$, because all probabilities are strictly between 0 and 1, the Lower Fréchet-Hoeffding Bound is:

$$C(u, v) = \max(u + v - 1, 0) \quad (2.5a)$$

Kendall's Tau

Kendall's tau can be computed for the Fréchet-Hoeffding Bounds, but due to the nature of these copulas, they are found easily. Knowing that Kendall's tau is simply the probability of concordance minus discordance as described in (1.2), and knowing that the Fréchet-Hoeffding Bounds are the extreme cases of exact concordance and exact discordance:

For the Upper Fréchet-Hoeffding Bound:

$$\tau = P(\text{Concordance}) - P(\text{Discordance})$$

$$P(\text{Discordance}) = 0$$

$$P(\textit{Concordance}) = 1$$

So,

$$\tau_{UFH} = 1 \quad (2.4b)$$

And for the Lower Fréchet-Hoeffding Bound:

$$\tau = P(\textit{Concordance}) - P(\textit{Discordance})$$

$$P(\textit{Discordance}) = 1$$

$$P(\textit{Concordance}) = 0$$

So,

$$\tau_{LFH} = -1 \quad (2.5b)$$

As expected, the Kendall's taus for the Upper and Lower Fréchet-Hoeffding bound are simply the upper bound and the lower bound of Kendall's tau itself respectively, representing both extreme cases of perfect concordance and perfect discordance.

Concordance Index

Finding the concordance indices of the Fréchet-Hoeffding bounds is just as intuitive as finding their Kendall's taus. By definition (1.4):

$$\kappa = P(\textit{Concordance}).$$

We know from the previous sections that the Upper and Lower Fréchet-Hoeffding bounds are the extreme cases of perfect concordance and discordance, respectively, so it is simple finding their concordance indices:

For the Upper Fréchet-Hoeffding Bound:

$$\kappa_{UFH} = 1 \quad (2.4c)$$

And for the Lower Fréchet-Hoeffding Bound:

$$\kappa_{LFH} = 0 \quad (2.5c)$$

Therefore, for the Upper Fréchet-Hoeffding Bound, any randomly chosen pair will be concordant, and for the Lower Fréchet-Hoeffding Bound, any randomly chosen pair will be discordant.

While the Fréchet-Hoeffding Bounds are not typically applied in models, they are extremely important theoretically, as they provide a space that all copulas will live between. They also serve as reference points for other copulas – other copulas can be understood as dependence structures that fall between these two extremes of perfect concordance and perfect discordance.

Gumbel Copula

Derivation

Similarly to the independence copula, the Gumbel Copula is a member of the Archimedean class of copulas. Therefore, the copula can be found with a generator function which is found directly from the Laplace transformation of the probability density function of a continuous non-negative random variable. In this case, the generator function and thus copula can be derived directly from the probability density function of a positive stable random variable:

Let $S \sim \text{Positive Stable}(\alpha)$ where $\alpha = \frac{1}{\theta}$, $1 \geq \alpha > 0$, $1 < \theta < \infty$. The pdf of a positive stable random variable is defined by its Laplace transformation:

$$\mathcal{L}(S) = \int_0^{\infty} e^{-tS} f_S(S) dS = e^{-t^\alpha}$$

So in this case, the inverse generator function can be found easily:

$$\phi_G^{-1}(t; \theta) = \exp\left\{-t^{\frac{1}{\theta}}\right\}, t \geq 0, \theta \geq 1$$

And therefore, the generator for the Gumbel copula can be found simply taking the inverse of the inverse:

$$t = \exp\left\{-\phi_G(t; \theta)^{\frac{1}{\theta}}\right\}$$

$$\phi_G = (-\ln(t))^\theta, t \geq 0, \theta \geq 1$$

Finally, for uniform random variables U and V by (2.1):

$$C(u, v) = \phi_G^{-1}(\phi_G(u) + \phi_G(v)) = \phi_G^{-1}((-\ln(u))^\theta + (-\ln(v))^\theta)$$

$$C(u, v) = \exp \left\{ - \left[(-\ln(u))^\theta + (-\ln(v))^\theta \right]^{\frac{1}{\theta}} \right\}, \quad t \geq 0, \theta \geq 1 \quad (2.6a)$$

The Gumbel copula has something that is new to copulas that have been derived so far: the parameter θ . This is called the *dependence parameter* and tweaking the value of the parameter will have different effects on the dependence structure of the two random variables. This parameter is especially useful in survival analysis frailty models. It will be discussed further later in this section when we derive the MLE of the Gumbel Copula.

Kendall's Tau

From the Copula definition and (1.3), Kendall's tau can be derived:

$$\tau = 4 \int_0^1 \int_0^1 C(u, v) dC(u, v) - 1$$

There is another useful definition that can be used to compute this value easier:

$$\tau = 1 + 4 \int_0^1 \frac{\phi(t)}{\phi'(t)} dt \quad (2.7)$$

We already have the generator for the Gumbel copula. Now, we have to find the derivative:

$$\phi'_G(t) = -\frac{\theta}{t} (-\ln(t))^{\theta-1}$$

And next, the quotient:

$$\frac{\phi_G(t)}{\phi'_G(t)} = \frac{(-\ln(t))^\theta}{-\frac{\theta}{t} (-\ln(t))^{\theta-1}}$$

$$\frac{\phi_G(t)}{\phi'_G(t)} = \frac{1}{\theta} t \ln(t)$$

Finally, we take the integral of this quantity from 0 to 1 which produces:

$$\tau = 1 + 4 \left[\frac{1}{2} t^2 \ln(t) - \frac{t^2}{4} \right]_{\{0\}}^1$$

Therefore, Kendall's tau for the Gumbel Copula is:

$$\tau_G = 1 - \frac{1}{\theta}, \theta \geq 1 \quad (2.6b)$$

Kendall's tau now depends on the dependence parameter, θ . Knowing that Kendall's tau is the probability of concordance minus the probability of discordance, this parameter θ tells us a few things about the Gumbel copula. Firstly, dependence is purely positive. Due to the bounds of the parameter θ , τ_G will always be between 0 and 1. Therefore, the probability of concordance will always be greater than the probability of discordance, except in the case where $\tau_G = 0$. This occurs when $\theta = 1$. When

$\theta = 1$, Gumbel's copula simplifies to the independence copula. $\tau_G = 0$ and it can even be seen in the formula for Gumbel's copula that setting $\theta = 1$ reduces the copula to independence:

$$C(u, v) = \exp\{ - [(-\ln(u))^1 + (-\ln(v))^1] \}$$

$$C(u, v) = uv$$

Furthermore, we can also observe the behavior as θ goes to infinity. As θ approaches infinity, the dependence term $\frac{1}{\theta}$ approaches 0, so τ_G approaches 1. Previously we saw that $\tau_G = 1$ is the definition of perfect concordance, which means that the Gumbel copula approaches the Upper Fréchet-Hoeffding bound as θ approaches infinity.

Concordance Index

The concordance index can be derived using (1.6):

$$\kappa = \frac{\tau + 1}{2}$$

Plugging in $1 - \frac{1}{\theta}$ for τ :

$$\kappa_G = \frac{1 - \frac{1}{\theta} + 1}{2}$$

$$\kappa_G = 1 - \frac{1}{2\theta}, \theta \geq 1 \quad (2.6c)$$

Once again, the concordance index depends on the parameter θ . Based on the bounds of θ , $\frac{1}{2} \leq \kappa_G \leq 1$. Therefore, the probability of concordance will always lie between 0.5 and 1 when modeling with the Gumbel copula. This means the Gumbel copula is limited to only modeling positive dependence. Particularly, Gumbel's copula is used for measuring upper tail dependence.

MLE

Since the Gumbel copula now directly involves a parameter θ , there must be a way to estimate this parameter when using real world data. If we have real data, we can fit a copula to deduce some information about the dependence pattern within the data. The process for estimation is typically as follows:

1. Start with observed pairs of data, say (x_i, y_i) , $i = 1, \dots, n$.
2. Transform this data into uniform data such that $(u_i, v_i) \in [0,1]$. This is typically done by transforming the data using the rank function:

$$u_i = \frac{\text{rank}(x_i)}{n+1}, v_i = \frac{\text{rank}(y_i)}{n+1}$$

3. Use the MLE to estimate the value of the dependence parameter, $\widehat{\theta}_{MLE}$.

Depending on the likelihood and the nature of $\widehat{\theta}_{MLE}$, this can be solved either numerically or analytically. But to even estimate numerically, we need to find the copula density function and then the likelihood function, and finally the MLE. Below is the full derivation for $\widehat{\theta}_{MLE}$:

First, define the copula density function:

$$c(u, v) = \frac{\partial^2}{\partial u \partial v} C(u, v) \quad (2.8)$$

Taking the first partial derivative with respect to u in the Gumbel Copula:

$$\begin{aligned} \frac{\partial}{\partial u} C(u, v) &= \exp \left\{ -[(-\ln(u))^\theta + (-\ln(v))^\theta]^{\frac{1}{\theta}} \right\} \cdot -\frac{1}{\theta} [(-\ln(u))^\theta + (-\ln(v))^\theta]^{\frac{1}{\theta}-1} \\ &\quad \cdot -\frac{\theta(-\ln(u))^{\theta-1}}{u} \end{aligned}$$

$$\text{Let } A = (-\ln(u))^\theta + (-\ln(v))^\theta$$

Simplifying:

$$\frac{\partial}{\partial u} C(u, v) = \exp\left\{-\left(A^{\frac{1}{\theta}}\right)\right\} \cdot -\frac{1}{\theta} A^{\frac{1}{\theta}-1} \cdot \frac{(-\ln(u))^{\theta-1}}{u}$$

Next, we can find the partial derivative with respect to v:

$$\text{Let } J = \frac{(-\ln(u))^{\theta-1}}{u} \leftarrow \text{const}$$

$$\begin{aligned} \frac{\partial^2}{\partial u \partial v} C(u, v) &= J \left[\exp\left\{-\left(A^{\frac{1}{\theta}}\right)\right\} \cdot \left(\frac{1}{\theta} - 1\right) [A]^{\frac{1}{\theta}-2} \cdot A'_v + \exp\left\{-\left(A^{\frac{1}{\theta}}\right)\right\} \cdot -\frac{1}{\theta} \left(A^{\frac{1}{\theta}-1} \cdot A'_v\right) \right. \\ &\quad \left. \cdot A^{\frac{1}{\theta}-1} \right] \end{aligned}$$

Finding A'_v :

$$A'_v = -\frac{\theta(-\ln(v))^{\theta-1}}{v}$$

Plugging in A'_v and simplifying:

$$\begin{aligned} \frac{\partial^2}{\partial u \partial v} C(u, v) &= J \left[\exp\left\{-\left(A^{\frac{1}{\theta}}\right)\right\} \cdot -\frac{\theta}{v} \left(\frac{1}{\theta} - 1\right) [A]^{\frac{1}{\theta}-2} \cdot (-\ln(v))^{\theta-1} \right] \\ &\quad + J \left[\exp\left\{-\left(A^{\frac{1}{\theta}}\right)\right\} \cdot -\left(A^{\frac{1}{\theta}-1} \cdot -\frac{1}{v} (-\ln(v))^{\theta-1}\right) \cdot A^{\frac{1}{\theta}-1} \right] \end{aligned}$$

Now, simplifying:

$$\begin{aligned} \frac{\partial^2}{\partial u \partial v} C(u, v) &= J \left[\exp\left\{-\left(A^{\frac{1}{\theta}}\right)\right\} \cdot (\theta - 1) [A]^{\frac{1}{\theta}-2} \cdot \frac{(-\ln(v))^{\theta-1}}{v} \right] \\ &\quad + J \left[\exp\left\{-\left(A^{\frac{1}{\theta}}\right)\right\} \cdot A^{\frac{2}{\theta}-2} \cdot \frac{(-\ln(v))^{\theta-1}}{v} \right] \end{aligned}$$

Factoring out the common terms:

$$\frac{\partial^2}{\partial u \partial v} C(u, v) = \frac{J \exp\left\{-\left(A^{\frac{1}{\theta}}\right)\right\} (-\ln(v))^{\theta-1}}{v} \left[(\theta - 1) A^{\frac{1}{\theta}-2} + A^{\frac{2}{\theta}-2} \right]$$

Finally, substituting in J and A gives the final copula density for the Gumbel Copula:

$$c(u, v) = \frac{(-\ln(u))^{\theta-1} (-\ln(v))^{\theta-1}}{uv} \exp\left\{-\left[(-\ln(u))^{\theta} + (-\ln(v))^{\theta}\right]\right\}$$

$$\cdot \left[(\theta - 1)[(-\ln(u))^\theta + (-\ln(v))^\theta]^{\frac{1}{\theta}-2} + [(-\ln(u))^\theta + (-\ln(v))^\theta]^{\frac{2}{\theta}-2} \right]$$

$$\theta \geq 1, u \in [0,1], v \in [0,1]. \quad (2.9a)$$

$\widehat{\theta}_{MLE}$ can be found by finding the maximum of the log likelihood function. Once again, I will define each part of the copula density function to make the computations easier.

Let

$$w = -\ln u \text{ and } x = -\ln v$$

$$A = w^\theta + x^\theta$$

Rewriting (2.9a):

$$c(u, v) = \frac{e^{-A^{\frac{1}{\theta}}} x^{\theta-1} y^{\theta-1}}{uv} \left[A^{\frac{2}{\theta}-2} + (\theta - 1) A^{\frac{1}{\theta}-2} \right]$$

Next, take the likelihood followed by the log likelihood.

$$L(\theta) = \prod_{i=1}^n \frac{e^{-A^{\frac{1}{\theta}}} x^{\theta-1} y^{\theta-1}}{uv} \left[A^{\frac{2}{\theta}-2} + (\theta - 1) A^{\frac{1}{\theta}-2} \right]$$

$$l(\theta) = \sum_{i=1}^n -A^{\frac{1}{\theta}} + (\theta - 1) \ln(x_i) + (\theta - 1) \ln(y_i) - \ln(u_i) - \ln(v_i)$$

$$+ \ln \left[A^{\frac{2}{\theta}-2} + (\theta - 1) A^{\frac{1}{\theta}-2} \right]$$

Now, take the derivative with respect to θ .

$$l'(\theta) = \sum_{i=1}^n \frac{\ln(A) \cdot A^{\frac{1}{\theta}}}{A^2} \cdot A' + \ln(x_i) + \ln(y_i) + \frac{A^{\frac{1}{\theta}-2} \left(\theta^2 - \ln(A)\theta - 2A^{\frac{1}{\theta}} \ln(A) + \ln(A) \right)}{A^{\frac{2}{\theta}-2} + (\theta-1)A^{\frac{1}{\theta}-2}} \quad (2.9b)$$

Clearly, this cannot be solved analytically for $\widehat{\theta}_{MLE}$. However, this is the furthest we can take the MLE method. Set the derivative with respect to θ of the log likelihood equal to 0 and solve for $\widehat{\theta}_{MLE}$. More concisely, we can write the MLE as:

$$\widehat{\theta}_{MLE} = \arg \max_{\theta \geq 1} (\sum_{i=1}^n \ln c(u_i, v_i)) \quad (2.9c)$$

The MLE is usually extremely analytically complicated for copulas. It might not always be easy to use this MLE. So, for a quicker trick, we can also just estimate $\hat{\theta}$ using Kendall's tau. This estimator provides a quick and convenient estimate, but it is not MVUE and certainly not the MLE. For the Gumbel copula, this estimator is:

$$\hat{\theta}_\tau = \frac{1}{1-\hat{\tau}} \quad (2.9d)$$

Clayton's Copula

Derivation

Clayton's copula is again an Archimedean copula and therefore can be derived easily with the proper probability density function. In this case, Clayton's copula can be derived using the Gamma distribution. Specifically, from the pdf of this random Gamma frailty variable:

$$S \sim \text{Gamma}\left(\frac{1}{\theta}, 1\right) \quad \theta > 0$$

Using the same methodology as the Independence and Gumbel copulas, we can find the inverse generator function by finding the Laplace transformation of the variable. The derivation follows:

$$\mathcal{L}(S; \theta) = E[e^{-tS}] = \int_0^\infty e^{-ts} f_s(s) ds$$

Where

$$f_s(s) = \frac{1}{\Gamma\left(\frac{1}{\theta}\right)} s^{\frac{1}{\theta}-1} e^{-s}, \quad s > 0, \quad \theta > 0$$

Then solving the integral:

$$\frac{1}{\Gamma\left(\frac{1}{\theta}\right)} \int_0^\infty s^{\frac{1}{\theta}-1} e^{-s(1+t)} ds$$

Recognizing this as the kernel of a $\text{Gamma}\left(\frac{1}{\theta}, \frac{1}{1+t}\right)$ distribution:

$$\frac{1}{\Gamma\left(\frac{1}{\theta}\right)} \cdot \Gamma\left(\frac{1}{\theta}\right) \left(\frac{1}{1+t}\right)^{\frac{1}{\theta}} \frac{1}{\Gamma\left(\frac{1}{\theta}\right) \left(\frac{1}{1+t}\right)^{\frac{1}{\theta}}} \int_0^\infty s^{\frac{1}{\theta}-1} e^{-s(1+t)} ds$$

The integral integrates out to 1 because it is a PDF. Therefore, the inverse of the Clayton generator is:

$$\phi_C^{-1} = (1 + t)^{-\frac{1}{\theta}}, \theta > 0, t \geq 0$$

And finding the generator function by taking this inverse:

$$\phi_C = t^{-\theta} - 1, \theta > 0, t \geq 0$$

Finally, to find the copula simply use (2.1) for random variables U and V:

$$C(u, v) = \phi_C^{-1}(\phi_C(u) + \phi_C(v))$$

$$C(u, v) = \phi_C^{-1}(u^{-\theta} + v^{-\theta} - 2)$$

$$C(u, v) = (u^{-\theta} + v^{-\theta} - 1)^{-\frac{1}{\theta}}, \theta > 0 \quad (3.10a)$$

Similarly to the Gumbel copula, Clayton's copula depends not only on the random variables U and V but also on the dependence parameter, θ .

Kendall's Tau

Derive Kendall's tau using definition (2.7), first by finding the derivative of the generator function:

$$\phi'_C(t) = -\theta t^{-\theta-1}$$

Then, find the integral:

$$\tau = 1 + 4 \int_0^1 -\frac{t^{-\theta} - 1}{\theta t^{-\theta-1}} dt$$

$$\tau = 1 - \frac{4}{\theta} \int_0^1 \frac{t^{-\theta}}{t^{-\theta} \cdot t^{-1}} - \frac{1}{t^{-\theta} t^{-1}} dt$$

$$\tau = 1 - \frac{4}{\theta} \int_0^1 t - t^{\theta+1} dt$$

$$\tau = 1 - \frac{4}{\theta} \left[\frac{t^2}{2} - \frac{1}{\theta+2} t^{\theta+2} \right]_{t_0}^{t_1}$$

$$\tau = 1 - \frac{2}{\theta} + \frac{4}{\theta(\theta + 2)}$$

$$\tau = \frac{\theta(\theta + 2)}{\theta(\theta + 2)} - \frac{2(\theta + 2)}{\theta(\theta + 2)} + \frac{4}{\theta(\theta + 2)}$$

$$\tau = \frac{\theta^2}{\theta(\theta + 2)}$$

$$\tau_C = \frac{\theta}{\theta + 2}, \theta > 0 \quad (3.10b)$$

Kendall's tau has some interesting properties for Clayton's copula. As θ approaches 0, Clayton's copula approaches the independence copula. Furthermore, as θ approaches ∞ , Clayton's copula approaches the Upper Fréchet-Hoeffding bound. Similarly to Gumbel's copula, Clayton's copula can only model positive dependence. However, there is a key difference. While Gumbel's copula is used to model upper tail positive dependence, Clayton's copula models lower tail positive dependence. Therefore, Gumbel's copula is suited well for variables that are concordant in the upper extremes while Clayton's copula is suited well for variables that are concordant in the lower extremes. This is due to how $C(u, v)$ behaves in the corners of its unit cube.

Concordance Index

The concordance index can be derived using (1.6):

$$\kappa = \frac{\tau + 1}{2}$$

$$\kappa = \frac{\frac{\theta}{\theta + 2} + 1}{2}$$

$$\kappa_C = \frac{\theta + 1}{\theta + 2}, \theta > 0 \quad (3.10c)$$

Similarly to Gumbel's copula, κ_C lives between $\frac{1}{2}$ and 1. Therefore, it can model only positive dependence. However, as explained previously, Clayton's copula is used to model purely lower tail dependence.

MLE

The MLE for Clayton's copula can be derived using the same methodology as for the Gumbel copula. Similarly to Gumbel, there is no analytical solution, but I will derive as far as I possibly can:

First, we must find the copula density function. From (3.10a) and (2.8):

$$C(u, v) = (u^{-\theta} + v^{-\theta} - 1)^{-\frac{1}{\theta}}$$

$$c(u, v) = \frac{\partial^2}{\partial u \partial v} C(u, v)$$

Finding the partial derivative with respect to u:

$$\frac{\partial}{\partial u} C(u, v) = u^{-\theta-1} (u^{-\theta} + v^{-\theta} - 1)^{-\frac{1}{\theta}-1}$$

Next, the partial derivative with respect to v:

$$\frac{\partial^2}{\partial u \partial v} C(u, v) = \left(\frac{1}{\theta} + 1\right) \frac{u^{-\theta-1} \cdot v^{-\theta-1}}{\theta} (u^{-\theta} + v^{-\theta} - 1)^{-\frac{1}{\theta}-2}$$

And therefore, the copula density function for Clayton's copula is:

$$c(u, v) = (\theta + 1)(uv)^{-(1+\theta)} (u^{-\theta} + v^{-\theta} - 1)^{-\frac{1}{\theta}-2}, \theta > 0, u \in [0,1], v \in [0,1] \quad (3.10d)$$

Finding the likelihood:

$$L(\theta) = \prod_{i=1}^n (\theta + 1)(u_i v_i)^{-(1+\theta)} (u_i^{-\theta} + v_i^{-\theta} - 1)^{-\frac{1}{\theta}-2}$$

Taking the log likelihood:

$$l(\theta) = \sum_{i=1}^n \ln(\theta + 1) - (1 + \theta)[\ln(u_i) + \ln(v_i)] - \left(\frac{1}{\theta} + 2\right) \ln(u_i^{-\theta} + v_i^{-\theta} - 1)$$

And finally, the partial derivative with respect to θ :

$$l'(\theta) = \sum_{i=1}^n \frac{1}{\theta+1} - \ln(u_i) - \ln(v_i) + \frac{1}{\theta^2} \ln(u_i^{-\theta} + v_i^{-\theta}) + \frac{\left(\frac{1}{\theta}+2\right)(\ln u_i \cdot u_i^{-\theta} + \ln v_i \cdot v_i^{-\theta-1})}{u_i^{-\theta} + v_i^{-\theta} - 1} \quad (3.10e)$$

$\widehat{\theta}_{MLE}$ will satisfy $l'(\theta) = 0$.

Similarly to Gumbel's copula, this must be solved numerically, and $\widehat{\theta}_{MLE}$ can be written:

$$\widehat{\theta_{MLE}} = \arg \max_{\theta > 0} \left(\sum_{i=1}^n \ln c(u_i, v_i) \right)$$

We can also derive a biased estimator of θ using the method of Kendall's tau. Using (3.10c):

$$\widehat{\theta_\tau} = \frac{2\hat{\tau}}{1-\hat{\tau}} \quad (3.10e)$$

Gaussian Copula

Derivation

The final copula I will be deriving is the Gaussian copula. The Gaussian copula is a very well-known copula that has faced lots of scrutiny, especially in the world of finance. The Gaussian copula has been credited as a major contributor to the financial crisis of 2008. The Gaussian copula was one of the models used to assess risk before the financial crisis. However, it was misused and misunderstood – the main reason being that it cannot model tail dependence. This means that it is not a great model for extreme events, and we will explore that through mathematical derivation.

Start with Sklar's theorem as defined in (1.1a)

$$H(x_1, x_2) = C(F_1(x_1), F_2(x_2))$$

Let $X = (Y_1, Y_2) \sim N(0, \Sigma)$ where Σ is the correlation matrix between (Y_1, Y_2) . (3.11)

Therefore, each marginal is $Y_i \sim N(0, \Sigma_{ii})$. And thus:

$$Y_1 \sim N(0, 1) \text{ and } Y_2 \sim N(0, 1).$$

Apply the probability integral transform to both random variables and say:

$$U_1 = \Phi(Y_1) \text{ and } U_2 = \Phi(Y_2)$$

Where Φ is the CDF of a standard normal. Because Φ is strictly monotone, it can be inverted and thus:

$$Y_1 = \Phi^{-1}(U_1) \text{ and } Y_2 = \Phi^{-1}(U_2)$$

Now, using the alternate definition of copula derived from Sklar's theorem (1.1b):

$$C(u_1, u_2) = P(U_1 \leq u_1, U_2 \leq u_2)$$

Substituting:

$$C(u_1, u_2) = P(\Phi(Y_1) \leq u_1, \Phi(Y_2) \leq u_2)$$

$$C(u_1, u_2) = P(Y_1 \leq \Phi^{-1}(u_1), Y_2 \leq \Phi^{-1}(u_2))$$

And this is simply the joint cdf of the multivariate normal distribution defined as X in (3.11). Therefore, the Gaussian copula can be written as this multivariate normal with the correlation matrix Σ .

$$C(u_1, u_2) = \Phi_{\Sigma}(\Phi^{-1}(u_1), \Phi^{-1}(u_2)) \quad u_1 \in [0,1] \quad u_2 \in [0,1] \quad (3.12a)$$

The Gaussian copula is the multivariate normal cdf. The only difference is that now the arguments of the cdf have been transformed into $Unif(0,1)$ which therefore satisfies Sklar's theorem.

Kendall's Tau

Kendall's tau can be found for the Gaussian copula by using equation (1.3):

$$\tau = 4 \int_0^1 \int_0^1 C(u, v) dC(u, v) - 1$$

This is the same thing as the expectation of $C(U_1, U_2)$:

$$\tau = 4E[C(U_1, U_2)] - 1$$

Converting back into $U_1 = \Phi(Y_1)$ and $U_2 = \Phi(Y_2)$;

$$\tau = 4E[C(\Phi(Y_1), \Phi(Y_2))] - 1$$

It follows directly from the Gaussian copula that:

$$\tau = 4E[\Phi_{\Sigma}(Y_1, Y_2)] - 1$$

Now to focus on the expectation:

$$\Phi_{\Sigma}(Y_1, Y_2) = P(Y'_1 < y_1, Y'_2 < y_2)$$

Where Y'_1 and Y'_2 are new independent copies from the same distribution. Now, condition on Y_1 and Y_2 :

$$\Phi_{\Sigma}(Y_1, Y_2) = P(Y'_1 < Y_1, Y'_2 < Y_2 \mid Y_1, Y_2)$$

Taking expectation of both sides yields:

$$E[\Phi_{\Sigma}(Y_1, Y_2)] = E[P(Y'_1 < Y_1, Y'_2 < Y_2 | Y_1, Y_2)]$$

So,

$$E[\Phi_{\Sigma}(Y_1, Y_2)] = P(Y'_1 \leq Y_1, Y'_2 \leq Y_2)$$

Let $D_1 = Y_1 - Y'_1$ and $D_2 = Y_2 - Y'_2$. Then,

$$\tau = 4P(D_1 \geq 0, D_2 \geq 0) - 1$$

By the orthant probability formula:

$$P(D_1 \geq 0, D_2 \geq 0) = \frac{1}{4} + \frac{1}{2\pi} \arcsin(\rho)$$

So therefore, Kendall's tau for the Gaussian is derived as:

$$\tau_{GAU} = \frac{2}{\pi} \arcsin(\rho), \quad -1 \leq \rho \leq 1 \quad (3.12b)$$

Notice that the dependence parameter is ρ in this case. When $\rho = -1$, the Gaussian copula becomes the Lower Fréchet-Hoeffding bound. When $\rho = 0$, it becomes the independence copula. Finally, when $\rho = 1$, it becomes the Upper Fréchet-Hoeffding bound. Therefore, the Gaussian copula can model negative dependence, independence, and positive dependence. However, the main drawback is that it cannot model upper or lower tail dependence due to the behavior of the underlying Gaussian structure of the copula. The Gaussian distribution has very weak tails, and this translates to the copula. This one misunderstanding is what many people consider the reason for the 2008 financial crisis.

Concordance Index

The concordance index of the Gaussian copula can be found easily from formula (1.6):

$$\kappa = \frac{\tau + 1}{2}$$

$$\kappa = \frac{\frac{2}{\pi} \arcsin(\rho) + 1}{2}$$

$$\kappa_{GAU} = \frac{1}{\pi} \arcsin(\rho) + \frac{1}{2}, \quad -1 \leq \rho \leq 1 \quad (3.12c)$$

The concordance index of the Gaussian copula confirms our findings with Kendall's tau. Since $-1 \leq \rho \leq 1$, it follows directly that $0 \leq \kappa_{GAU} \leq 1$. Thus, the Gaussian copula can model all types of dependence between two continuous random variables.

MLE

The MLE for the Gaussian copula can be derived in the same way it was derived for the Gumbel copula and Clayton's copula. First, we must identify the copula density function, and from there we can use the likelihood function to find the MLE. The following derivation first defines the copula density function and then the MLE:

Start with the definition of the Gaussian copula:

$$C(u_1, u_2) = \Phi_{\Sigma}(\Phi^{-1}(u_1), \Phi^{-1}(u_2)) \quad u_1 \in [0,1] \quad u_2 \in [0,1]$$

Now, let $z_1 = \Phi^{-1}(u_1)$ and $z_2 = \Phi^{-1}(u_2)$. Thus, we can rewrite the Gaussian copula:

$$C(u_1, u_2) = \Phi_{\Sigma}(z_1, z_2)$$

Therefore, the Gaussian copula is the gaussian multivariate cumulative distribution function evaluated at z_1 and z_2 . With this information we can simply find multivariate gaussian probability distribution function evaluated at z_1 and z_2 . However, we need to remove information about the marginals so only the copula's dependence information remains. So, by Sklar's theorem, we can write the Gaussian copula density function as:

$$c(z_1, z_2) = \frac{f_{z_1, z_2}(z_1, z_2)}{\phi(z_1)\phi(z_2)}$$

Where $f_{z_1, z_2}(z_1, z_2)$ is the multivariate Gaussian pdf and $\phi(z_1)\phi(z_2)$ is the marginal density of the two standard normal variables multiplied by one another. Now, substituting the known values and since $Z_1 \sim N(0,1)$ and $Z_2 \sim N(0,1)$:

$$c(z_1, z_2) = \frac{\frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2[1-\rho^2]}[z_1^2 - 2\rho z_1 z_2 + z_2^2]\right\}}{\frac{1}{2\pi} \exp\left\{-\frac{1}{2}(z_1^2 + z_2^2)\right\}}$$

Which simplifies to:

$$c(z_1, z_2) = \frac{1}{\sqrt{(1-\rho^2)}} \exp\left\{-\frac{1}{2[1-\rho^2]}[z_1^2 - 2\rho z_1 z_2 + z_2^2] + \frac{1}{2}[z_1^2 + z_2^2]\right\} \quad 3.12(d)$$

Now the MLE can be found by maximizing the likelihood as such:

$$L(\rho) = \prod_{i=1}^n \frac{1}{\sqrt{(1-\rho^2)}} \exp \left\{ -\frac{1}{2[1-\rho^2]} [z_{1i}^2 - 2\rho z_{1i}z_{2i} + z_{2i}^2] + \frac{1}{2} [z_{1i}^2 + z_{2i}^2] \right\}$$

Taking the log likelihood:

$$\begin{aligned} l(\rho) &= \sum_{i=1}^n \ln(1) - \frac{1}{2} \ln((1-\rho^2)) - \frac{1}{2[1-\rho^2]} [z_{1i}^2 - 2\rho z_{1i}z_{2i} + z_{2i}^2] + \frac{1}{2} [z_{1i}^2 + z_{2i}^2] \\ l(\rho) &= -\frac{n}{2} \ln((1-\rho^2)) - \frac{1}{2[1-\rho^2]} \left[\sum_{i=1}^n [z_{1i}^2 + z_{2i}^2] - \sum_{i=1}^n 2\rho z_{1i}z_{2i} \right] + \frac{1}{2} \sum_{i=1}^n [z_{1i}^2 + z_{2i}^2] \end{aligned}$$

Let $A = \sum_{i=1}^n [z_{1i}^2 + z_{2i}^2]$ and $B = \sum_{i=1}^n z_{1i}z_{2i}$. Now rewriting the log likelihood:

$$l(\rho) = -\frac{n}{2} \ln((1-\rho^2)) - \frac{1}{2[1-\rho^2]} [A - 2\rho B] + \frac{1}{2} A$$

Taking the derivative with respect to ρ :

$$l'(\rho) = \frac{n\rho}{1-\rho^2} + \frac{B\rho^2 - A\rho + B}{(\rho^2 - 1)^2}$$

This can actually be solved analytically. However, in practice this is just solved numerically. For the sake of this project, I will solve as much as I can:

$$0 = \frac{n\rho}{1-\rho^2} + \frac{B\rho^2 - A\rho + B}{(\rho^2 - 1)^2}$$

$$n\rho = \frac{B\rho^2 - A\rho + B}{\rho^2 - 1}$$

$$n\rho^3 - n\rho = B\rho^2 - A\rho + B$$

$$n\rho^3 - B\rho^2 + (A - n)\rho - B = 0 \quad (3.12e)$$

This cubic actually does have an analytical solution using Cardano's formula. But it is extremely messy and complicated. Therefore, the value of ρ which causes this equation to be 0 is

$$\widehat{\rho_{MLE}}.$$

Another simple biased estimator is found from Kendall's tau:

$$\widehat{\rho} = \sin\left(\frac{\pi}{2}\hat{\tau}\right) \quad (3.12f)$$

IV. Results

The results of all the derivations are condensed into the tables below:

Copula	$C(u, v)$	$c(u, v)$	$\phi(t)$	$\phi^{-1}(t)$
Independence	uv	Not derived	$-\ln(t)$	e^{-t}
Fréchet-Hoeffding Lower Bound	$\max(u + v - 1, 0)$	Not derived	N/A	N/A
Fréchet-Hoeffding Upper Bound	$\min(u, v)$	Not derived	N/A	N/A

Copula	Kendall's tau	Concordance Index	MLE
Independence	$\tau_I = 0$	$\kappa_I = \frac{1}{2}$	N/A
Fréchet-Hoeffding Lower Bound	$\tau_I = -1$	$\kappa_I = 0$	N/A
Fréchet-Hoeffding Upper Bound	$\tau_I = 1$	$\kappa_I = 1$	N/A

Copula	$C(u, v)$	$c(u, v)$
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Gumbel	$\exp \left\{ - \left[(-\ln(u))^{\theta} + (-\ln(v))^{\theta} \right]^{\frac{1}{\theta}} \right\}$	See below*
Clayton	$(u^{-\theta} + v^{-\theta} - 1)^{-\frac{1}{\theta}}$	$(\theta + 1)(uv)^{-(1+\theta)}(u^{-\theta} + v^{-\theta} - 1)^{\frac{1}{\theta}-2}$
Gaussian	$\Phi_{\Sigma}(\Phi^{-1}(u_1), \Phi^{-1}(u_2))$	See below**

*Gumbel copula density: $c(u, v) = \frac{(-\ln(u))^{\theta-1}(-\ln(v))^{\theta-1}}{uv} \exp \left\{ - \left[(-\ln(u))^{\theta} + (-\ln(v))^{\theta} \right] \right\} \cdot \left[(\theta - 1) \left[(-\ln(u))^{\theta} + (-\ln(v))^{\theta} \right]^{\frac{1}{\theta}-2} + \left[(-\ln(u))^{\theta} + (-\ln(v))^{\theta} \right]^{\frac{2}{\theta}-2} \right]$

**Gaussian copula density: $\frac{1}{\sqrt{(1-\rho^2)}} \exp \left\{ -\frac{1}{2[1-\rho^2]} [z_1^2 - 2\rho z_1 z_2 + z_2^2] + \frac{1}{2} [z_1^2 + z_2^2] \right\}$

Copula	$\phi(t)$	$\phi^{-1}(t)$
Gumbel	$(-\ln(t))^{\theta}$	$\exp \left\{ -t^{\frac{1}{\theta}} \right\}$
Clayton	$t^{-\theta} - 1$	$(1 + t)^{-\frac{1}{\theta}}$
Gaussian	N/A	N/A

Copula	Kendall's tau	Concordance Index	MLE
Gumbel	$\tau_G = 1 - \frac{1}{\theta}, \theta \geq 1$	$\kappa_G = 1 - \frac{1}{2\theta}, \theta \geq 1$	$\hat{\theta}_{MLE} = \arg \max_{\theta \geq 1} \left(\sum_{i=1}^n \ln c(u_i, v_i) \right)$
Clayton	$\tau_C = \frac{\theta}{\theta + 2}, \theta > 0$	$\kappa_C = \frac{\theta + 1}{\theta + 2}, \theta > 0$	$\hat{\theta}_{MLE} = \arg \max_{\theta \geq 0} \left(\sum_{i=1}^n \ln c(u_i, v_i) \right)$
Gaussian	$\tau_{GAU} = \frac{2}{\pi} \arcsin(\rho)$	$\frac{1}{\pi} \arcsin(\rho) + \frac{1}{2}$	$n\rho^3 - B\rho^2 + (A - n)\rho - B = 0^{***}$

***Solve numerically or with Cardano's formula

V. Conclusion

Copulas provide a powerful way to capture the full dependence structure between two random variables. Instead of inspecting raw variables manually, we transformed them into their respective rank or percentile scale, producing new uniform random variables. The copula then describes with a mathematical function how these two variables move together in rank.

In this project, we examined multiple different copulas including their dependence measures, Kendall's tau and the concordance index. Starting with Sklar's theorem, we were able to derive these copulas and their respective measures. These derivations show how vastly different copula structures can be, even when marginals are the same. For example, Gaussian copulas can model symmetric dependence but not tail dependence. On the other hand, Clayton's copula models positive dependence in the lower tail while Gumbel's copula excels at modeling positive dependence in the upper tail. There is a potentially infinite number of copulas, all of which are unique and explain a specific dependence structure.

The main takeaway from this project is that the dependence between two random variables cannot always just be described by correlation. While typical measures such as Pearson's correlation coefficient can model linear normal data on average, copulas can describe the complete dependence structure. The data can be linear or nonlinear, and complex dependencies such as those within the tails of the distribution can be accounted for. Measures such as Kendall's tau and the concordance index are especially useful because they describe dependence through rank association rather than through raw numerical values, allowing us to better capture these complex dependencies.

Overall, copulas are valuable because they provide a systematic way to construct joint distributions from separate marginal distributions while preserving detailed information about dependence. Their ability to model rank dependence, nonlinear dependence, asymmetric dependence, and tail behavior makes them extremely useful in many applied areas such as finance, survival analysis, insurance, environmental modeling, and risk analysis. Through this project, copulas were shown to be not only a mathematical tool, but also as a tool to better understand and model how variables move together.

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